PhishNet v2

Download and Installation Manual

Extension version 2.3.0

Public release edition

Group 11

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https://github.com/aaron-c-s-ramos/PhishNet-v2-Public-Release

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1 About this manual

This manual gives the procedures to download, install, update, and remove the PhishNet v2 browser extension. It is applicable to version 2.3.0.

The manual is for persons who install PhishNet on a personal computer. The procedures do not use programming tools, a command terminal, or a GitHub account.

Conventions

Convention	Meaning
Bold text	A label, a button, a tab, or a switch on the screen.
`Monospace text`	Text that you type, a file name, a folder name, or an address.
Result:	What the screen shows when the step is correct.
CAUTION	A risk of damage to the installation or a risk to your security.
NOTE	Information that helps you. A note does not contain an instruction.

2 About PhishNet v2

PhishNet v2 is an anti-phishing extension for Chromium browsers and for Firefox. It uses Manifest V3. It examines the address of each web page that you open.

PhishNet examines an address in two steps:

- First, PhishNet compares the address to five public threat feeds and to the community blacklist. If a list contains the address, PhishNet stops the page before it opens. A warning page shows the name of the list.
- If no list contains the address, an on-device Random Forest classifier calculates a risk score from the address. If the score is in the Caution band, a warning shows on top of the page. The classifier alone never stops a page.

PhishNet does all checks on your computer. It does not send an address to a server for a check. It does not use an account or an API key. Protection starts immediately after installation.

Community reporting is optional and is off in this release. Section 8 gives the details. A report contains only a hashed installation identifier, the address, and the time.

3 Before you start

Item	Requirement
Browser	Google Chrome, Microsoft Edge, Brave, Opera, or Vivaldi, version 116 or later. Or Firefox 140 or later (Procedure E).
Operating system	Windows, macOS, or Linux.
Internet connection	Necessary for the download and for the first download of the threat feeds. The PhishTank feed alone is approximately 14 MB.

Item	Requirement
Disk space	Approximately 22 MB: 4.4 MB for the ZIP file and 17 MB for the extracted folder.
Account	Not necessary. The repository is public, and PhishNet uses no account.

> **NOTE**
> Firefox users do Procedure E in place of procedures A, B, and C. Then they do Procedure D.

Do the procedures in sequence: A, B, C, and then D.

4 Procedure A: Download PhishNet

Use one of the two methods below. Method 1 is recommended.

Method 1: Download from the Releases page

1. Open the Releases page:
   ```
   https://github.com/aaron-c-s-ramos/PhishNet-v2-Public-Release/releases
   ```
2. Find the newest release at the top of the page.
   Result: At the date of this manual, the newest release is **PhishNet v2.3.0**.
3. Under **Assets**, click `phishnet-2.3.0.zip` to download it.
   Result: The browser downloads a file of approximately 4.4 MB.
4. Do not download **Source code (zip)** or **Source code (tar.gz)**. These files contain the repository, not the extension folder.

Method 2: Download from the file list

1. Open the repository page:
   ```
   https://github.com/aaron-c-s-ramos/PhishNet-v2-Public-Release
   ```
2. In the file list, click `phishnet-2.3.0.zip`.
3. Click **Download raw file**. This button is at the top right of the file view.
   Result: The browser downloads a file of approximately 4.4 MB.

Extract the ZIP file

1. Make a folder for the extension in a location that you will keep. For example:
   - Windows: `C:\PhishNet`
   - macOS or Linux: `~/PhishNet`
2. Extract the contents of `phishnet-2.3.0.zip` into that folder.
   - Windows: right-click the file, click **Extract All**, then select the folder.
   - macOS: double-click the file, then move the contents into the folder.
   - Linux: use your archive manager or the `unzip` command.

3. Open the folder. Make sure that `manifest.json` is directly in the folder.

   Result: The folder contains `assets`, `block.html`, `fonts`, `icons`, `manifest.json`, `model`, `options.html`, `ort`, `popup.html`, and `rules`.

> **CAUTION**
> Do not move, rename, or delete this folder after the installation. The browser loads the extension from this folder each time it starts. If the folder is not there, PhishNet stops.

Go to Procedure B.

5 Procedure B: Load the extension in the browser

The steps are the same in all Chromium browsers. Only the address of the extensions page is different.

Browser	Extensions page address
Google Chrome	`chrome://extensions`
Microsoft Edge	`edge://extensions`
Brave	`brave://extensions`
Opera	`opera://extensions`
Vivaldi	`vivaldi://extensions`

1. Open the browser.
2. Type the extensions page address in the address bar, then push Enter.
3. Set **Developer mode** to on.
   - Chrome, Brave, Opera, and Vivaldi: the switch is at the top right of the page.
   - Edge: the switch is in the left panel.

   Result: The **Load unpacked** button shows.
4. Click **Load unpacked**.
5. Select the folder that you made in Procedure A, for example `C:\PhishNet`. Do not select a file in the folder.

   Result: The **PhishNet** card shows version **2.3.0** and no error.
6. Click the extensions icon in the browser toolbar. In Chrome and Edge, this icon is a puzzle piece.
7. Click the pin icon next to **PhishNet**.

   Result: The PhishNet icon stays in the toolbar.

> **NOTE**
> The browser can show a message about extensions in developer mode. This message is usual for an extension that you load from a folder.

> **NOTE**
> PhishNet uses permissions to examine each address before the page loads. Appendix A gives each permission and its purpose.

6 Procedure C: Turn on protection in private windows

> **CAUTION**
> Set the private-window switch to on. If the switch is off, PhishNet does not examine the pages that you open in private windows.

The browser keeps PhishNet off in private windows by default. Only you can change this setting. PhishNet cannot change it.

1. Open the extensions page of the browser.
2. On the **PhishNet** card, click **Details**.
3. Set the private-window switch to on:
   - Chrome: **Allow in Incognito**
   - Edge: **Allow in InPrivate**
   - Brave, Opera, and Vivaldi: the switch for private windows. The label can be different in each browser.
4. Open the PhishNet settings. Click the PhishNet icon, then click the gear icon.

   Result: Under **Private browsing**, the setting shows `granted`. Figure 3 shows this setting before the switch is on.

> **NOTE**
> In a private window, PhishNet keeps its data in session storage. The browser deletes this data when the private session closes.

7 Procedure D: Do a check of the installation

1. Click the PhishNet icon in the toolbar.

   Result: The PhishNet popup opens on the **This page** tab (Figure 1, left).
2. Click the **Threat feeds** tab.

   Result: The first download of the feeds starts automatically (Figure 1, right).
3. Wait approximately one minute. The PhishTank feed alone is approximately 14 MB.
4. Under **Sources**, make sure that each of the five feeds shows a number of **entries**.
   - OpenPhish
   - ValidIn
   - PhishTank
   - PhishStats

- URLhaus

5. If a feed shows no entries, click **Refresh now**, then wait one more minute.
6. Find the **On-device classifier** panel on the same tab.

   Result: The **model** row shows `random forest 20261016-012038`. The **corpora** row shows `ramyasree, hannousse, phiusiil`.
7. Optional: type an address in **Check an address**, then click **Check**.

   Result: PhishNet shows the risk band of that address. PhishNet does not open the page.

> **NOTE**
>
> After the first download, PhishNet downloads the feeds again every 24 hours. You can change this interval in the settings to 6, 12, or 48 hours.

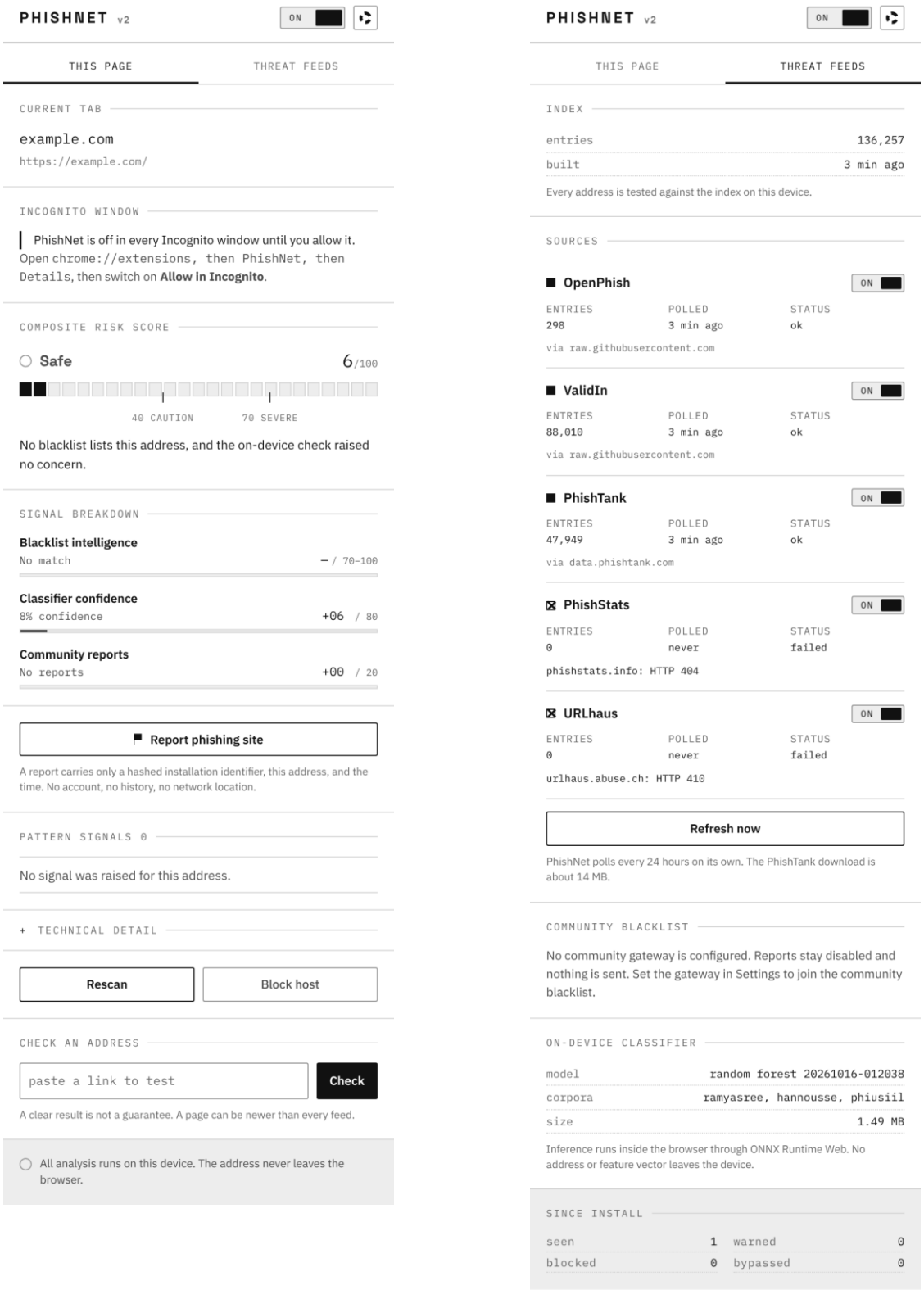


Figure 1. The PhishNet popup. Left: the **This page** tab for `https://example.com/`. Right: the **Threat feeds** tab.

> **NOTE**
>
> In this capture, the network of the capture computer blocked PhishStats and URLhaus. Thus these two feeds show `failed`. On a usual connection, the five feeds show entries.

> **NOTE**
>
> The **Incognito window** panel on the **This page** tab shows only when the private-window switch is off. In this capture, the switch was off.

Installation checklist

	Item
☐	The **PhishNet** card shows version **2.3.0** and no error.
☐	The PhishNet icon is pinned to the toolbar.
☐	The private-window switch is on, and **Private browsing** shows `granted`.
☐	The **Threat feeds** tab shows entries for the five feeds.
☐	The **On-device classifier** panel shows the random forest model.

The installation is complete. The **This page** tab shows the risk score and the signals of the current page. The gear icon opens **Settings**.

The Settings page

The **Settings** page keeps each change immediately. It has no save button, except for the gateway origin. Figures 2 and 3 show the page with its default values.

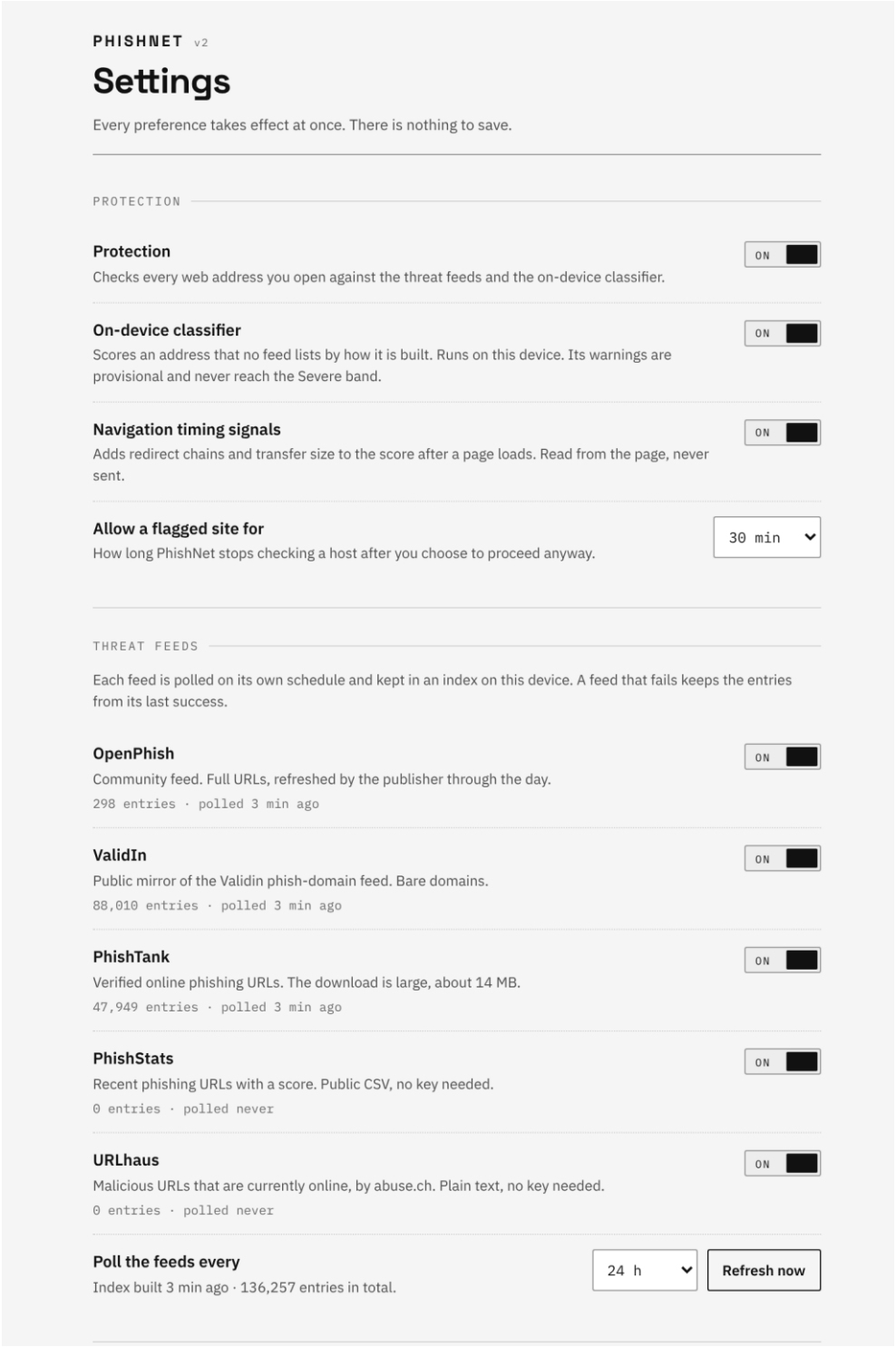


Figure 2. The Settings page, upper part: **Protection** and **Threat feeds**.

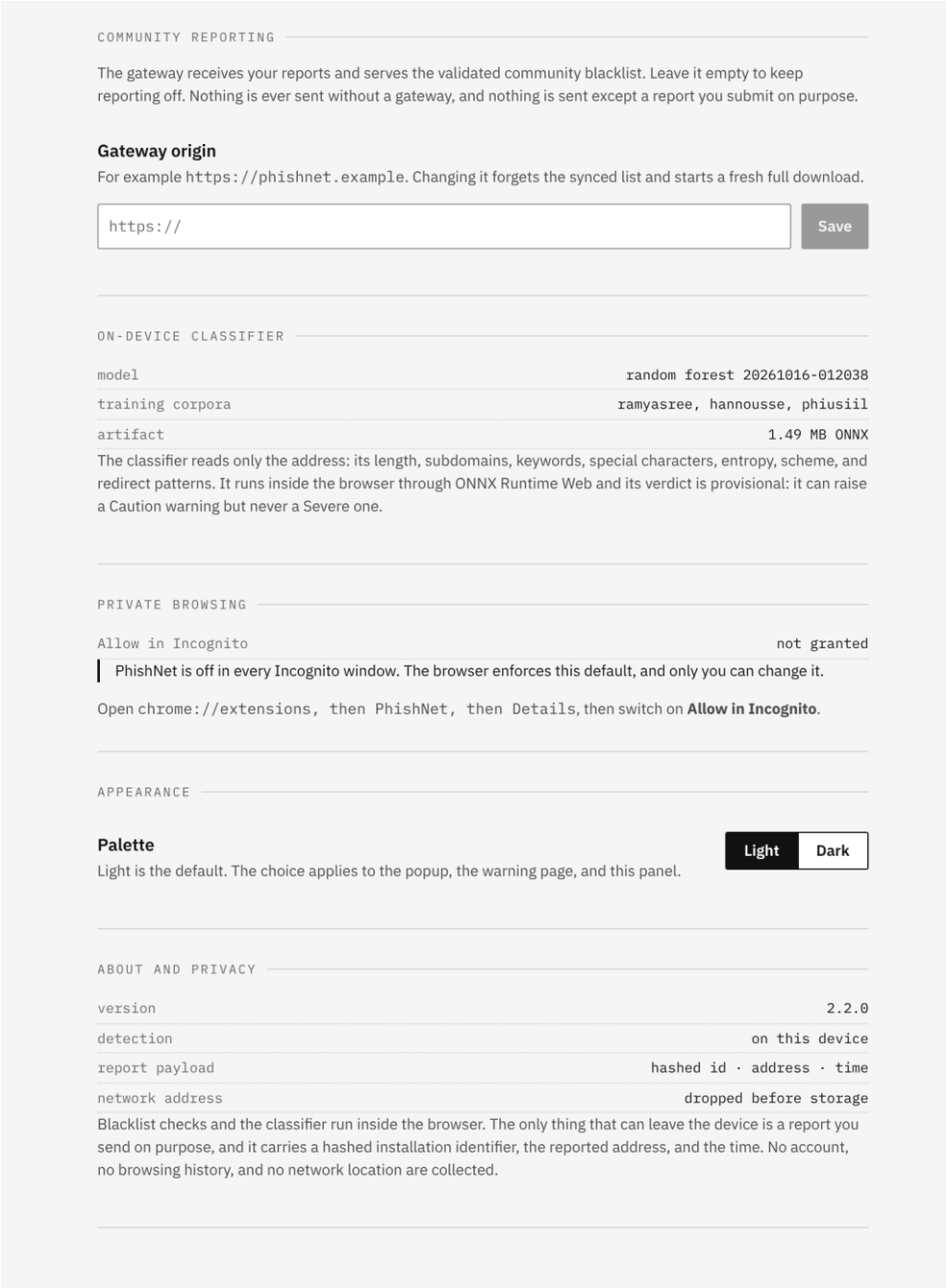


Figure 3. The Settings page, lower part: **Community reporting**, **On-device classifier**, **Private browsing**, **Appearance**, and **About and privacy**.

> **NOTE**
> In this capture, the private-window switch was off. Thus **Allow in Incognito** shows `not granted`.

8 Procedure E: Install PhishNet in Firefox

Firefox 140 or later uses a separate package. Mozilla signed this package, thus Firefox keeps PhishNet after a restart.

1. Open the Releases page:

```
https://github.com/aaron-c-s-ramos/PhishNet-v2-Public-Release/releases
```

2. Under **Assets**, click `phishnet-2.3.0-firefox.xpi` to download it.

   Result: The browser downloads a file of approximately 4.4 MB.

3. In Firefox, type this address in the address bar, then push Enter:

   ```
   about:addons
   ```

4. Click the gear icon, then click **Install Add-on From File**.

5. Select the file `phishnet-2.3.0-firefox.xpi`.

   Result: Firefox asks if you want to add PhishNet.

6. Click **Add**.

   Result: PhishNet shows in the list of extensions.

7. Click **PhishNet**, then click the **Details** tab.

8. Set **Run in Private Windows** to **Allow**.

9. Click the extensions icon in the toolbar. Click the gear next to **PhishNet**, then click **Pin to Toolbar**.

10. Do Procedure D.

> **NOTE**
>
> Browsers that use the Firefox engine, for example LibreWolf, Waterfox, Floorp, and Zen, can read the same package. The project team did not do a test of these browsers.

> **NOTE**
>
> Firefox asks for your consent before the first community report. Without a gateway, PhishNet sends nothing (section 9).

9 Community reporting

The **Report phishing site** action and the shared community blacklist need a community gateway. A community gateway is a server that receives reports. This public release does not include a gateway.

Without a gateway, PhishNet sends nothing. The report action shows "No community gateway is configured or reachable". All other checks operate fully.

If the project team gives you a gateway address, connect PhishNet to it:

1. Click the PhishNet icon, then click the gear icon.

2. Find the **Community reporting** section (Figure 3).

3. In **Gateway origin**, type the gateway address that you received.

4. Click **Save**.

   Result: The section shows the **admission threshold** and the time of the **last sync**.

> **CAUTION**
>
> Type only a gateway address that the project team gave you. A report goes to the gateway that you type in **Gateway origin**.

10 Update PhishNet

The browser identifies an extension that you load from a folder by the path of the folder. Use the same folder for each update. Then PhishNet keeps its settings.

1. Download the newer ZIP file from the Releases page (Procedure A).
2. Delete all files in your PhishNet folder. Do not delete the folder.
3. Extract the contents of the newer ZIP file into the same folder.
4. Open the extensions page of the browser.
5. On the **PhishNet** card, click the reload icon (a circular arrow).
6. Make sure that the card shows the new version number.
7. Make sure that the private-window switch is still on.

> **NOTE**
> If you delete the old files first, no old files stay in the folder. The file names in the `assets` folder change with each release.

11 Remove PhishNet

1. Open the extensions page of the browser.
2. On the **PhishNet** card, click **Remove**. Then click **Remove** again in the dialog box.
3. Delete the PhishNet folder and the ZIP file.

> **NOTE**
> The browser deletes the settings and the threat-feed index of PhishNet when you remove the extension.

12 Troubleshooting

Problem	Possible cause	Remedy
The **Load unpacked** button does not show.	**Developer mode** is off.	Set **Developer mode** to on (Procedure B, step 3).
The browser shows "Manifest file is missing or unreadable".	The selected folder does not contain `manifest.json` directly.	Select the folder that contains `manifest.json` (Procedure A, "Extract the ZIP file", step 3).
The folder contains `README.md`, the manual files, and `phishnet-2.3.0.zip`, but no `manifest.json`.	You downloaded **Source code (zip)** and not `phishnet-2.3.0.zip`.	Download `phishnet-2.3.0.zip` (Procedure A, Method 1, step 3). Then do "Extract the ZIP file" again.
PhishNet shows an error or is gone after a restart of the browser.	The PhishNet folder was moved, renamed, or deleted.	Put the folder back, or load the extension again from the new location.
Private windows show no PhishNet warnings.	The private-window switch is off.	Do Procedure C.
A feed shows no entries after one minute.	The download is not complete, or a firewall blocks the feed server.	Click **Refresh now**. Make sure that your network does not block the servers in the note below this table.
One feed shows an error. The other feeds show entries.	The server of that feed did not answer.	No action is necessary. PhishNet keeps the entries of the last successful download and tries again later.
The **model** row shows `not loaded` and **fallback** shows `pattern heuristics`.	The classifier did not start.	Reload PhishNet on the extensions page. If the problem continues, download and extract the ZIP file again.
Report phishing site shows "No community gateway is configured or reachable".	No gateway origin is saved. This release has no gateway.	No action is necessary. Refer to section 9.
Firefox does not accept the file.	Firefox is older than version 140, or you selected the Chromium ZIP file.	Update Firefox. Install `phishnet-2.3.0-firefox.xpi` (Procedure E).
The PhishNet window shows **Website access**.	The browser does not let PhishNet read websites.	Click **Allow access to websites**, then accept the browser prompt.

NOTE

PhishNet downloads the threat feeds from these servers: `raw.githubusercontent.com`, `openphish.com`, `data.phishtank.com`, `phishstats.info`, and `urlhaus.abuse.ch`.

Appendix A: Permissions

The browser can show this list of permissions at the installation. PhishNet uses each permission only for the purpose below.

Permission	Purpose
`storage`	Keeps the threat-feed index, the settings, and the community list on your computer.
`tabs`	Finds the address of the current tab for the popup.
`alarms`	Starts the hourly check of the feeds and the community list.
`scripting`	Reads the Navigation Timing data of a page after the page loads.
`webNavigation`	Gets the address of a page before the page loads.
`declarativeNetRequest`	Stops a listed page at the network level.
`declarativeNetRequestFeedback`	Identifies which rule stopped a page.
Access to all `http` and `https` sites	Examines each address that you open, and shows the warning over a page.

Appendix B: Glossary

Term	Meaning
Unpacked extension	An extension that the browser loads from a folder on your computer, and not from a web store.
Developer mode	A setting of the extensions page that lets you load an unpacked extension.
Threat feed	A public list of known phishing or malicious addresses.
Community blacklist	A list of addresses that PhishNet users reported and that a community gateway accepted.
Community gateway	A server that receives reports and sends the community blacklist to the extension.
Classifier	The machine learning model that calculates a risk score from an address.
Gateway origin	The address of a community gateway, for example `https://phishnet.example`.